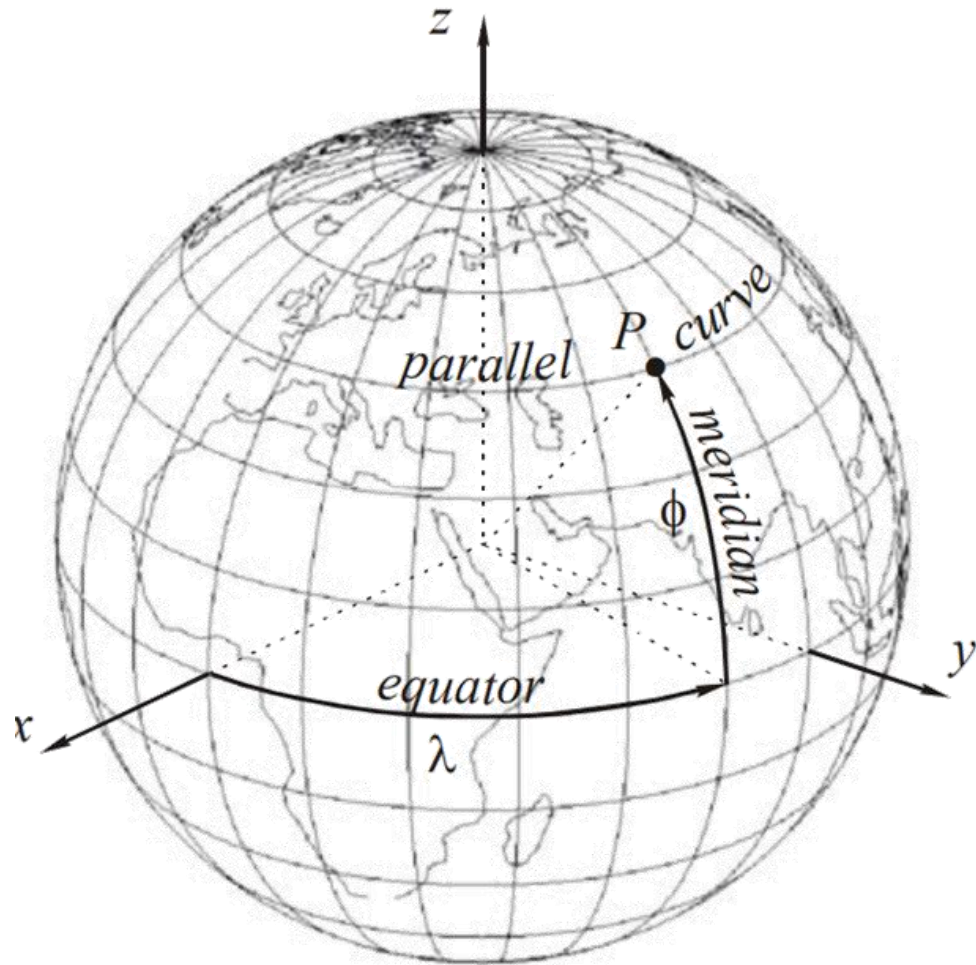
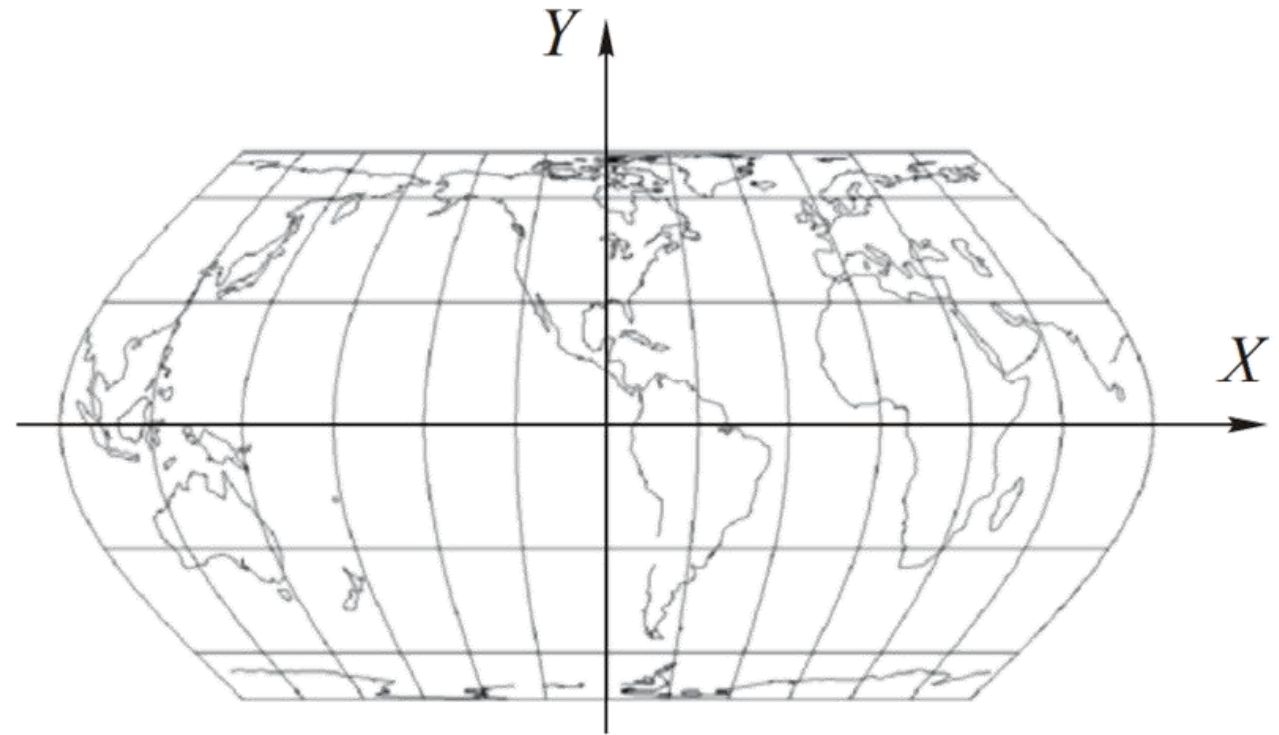


Map Projection Theory

Geographic Coord.



Projected Coord.



x, y, z Cartesian vs ϕ, λ curvilinear

$$x = f_1(\phi, \lambda) = R \cos \phi \cos \lambda$$

$$y = f_2(\phi, \lambda) = R \cos \phi \sin \lambda$$

$$z = f_3(\phi, \lambda) = R \sin \phi$$

map projection

$$X = F_1(U, V)$$

$$Y = F_2(U, V)$$

$$Z = F_3(U, V)$$

$$X = \bar{f}_1(\phi, \lambda)$$

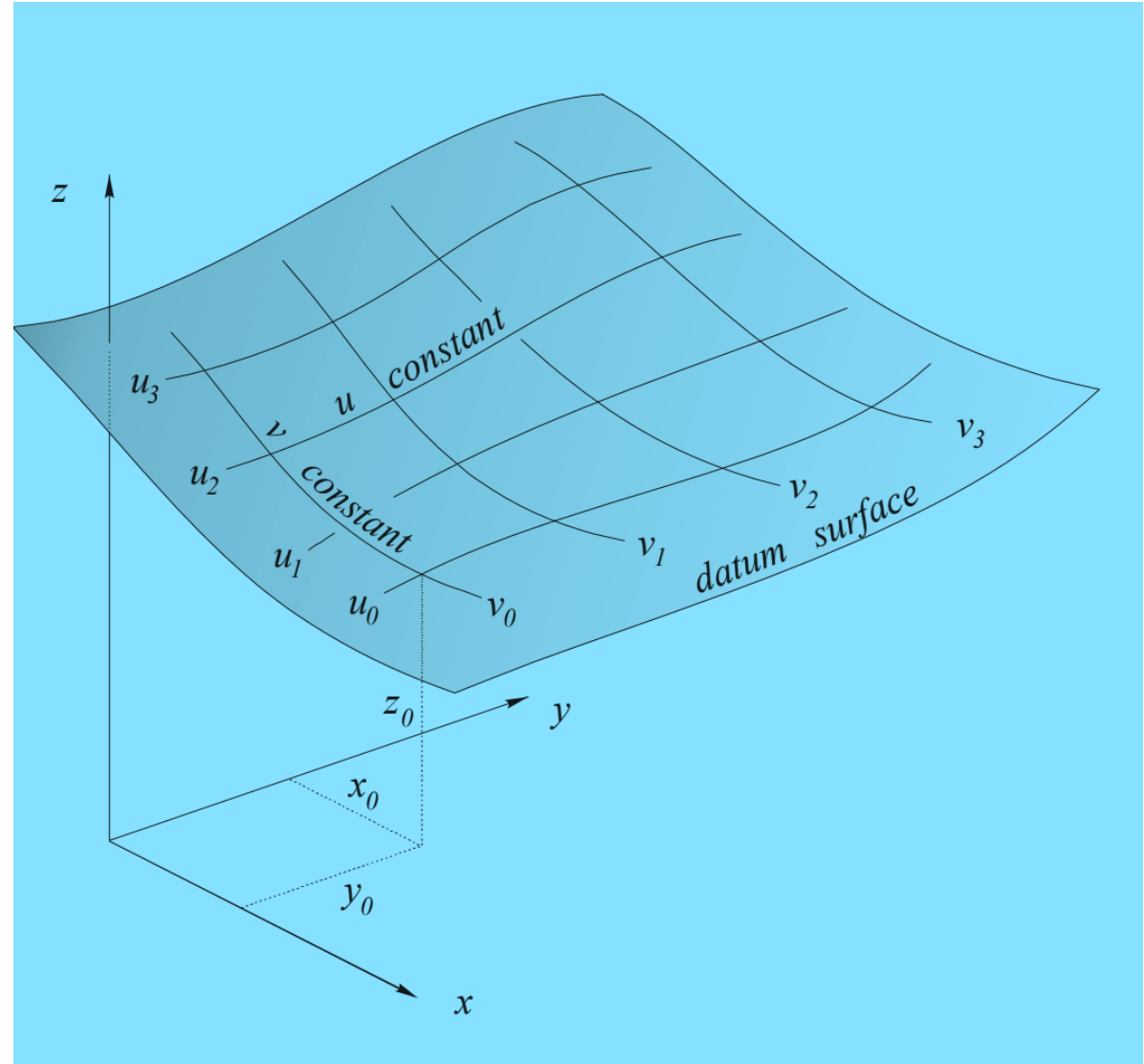
$$Y = \bar{f}_2(\phi, \lambda)$$

Datum Surface

$$x = f_1(u, v)$$

$$y = f_2(u, v)$$

$$z = f_3(u, v)$$

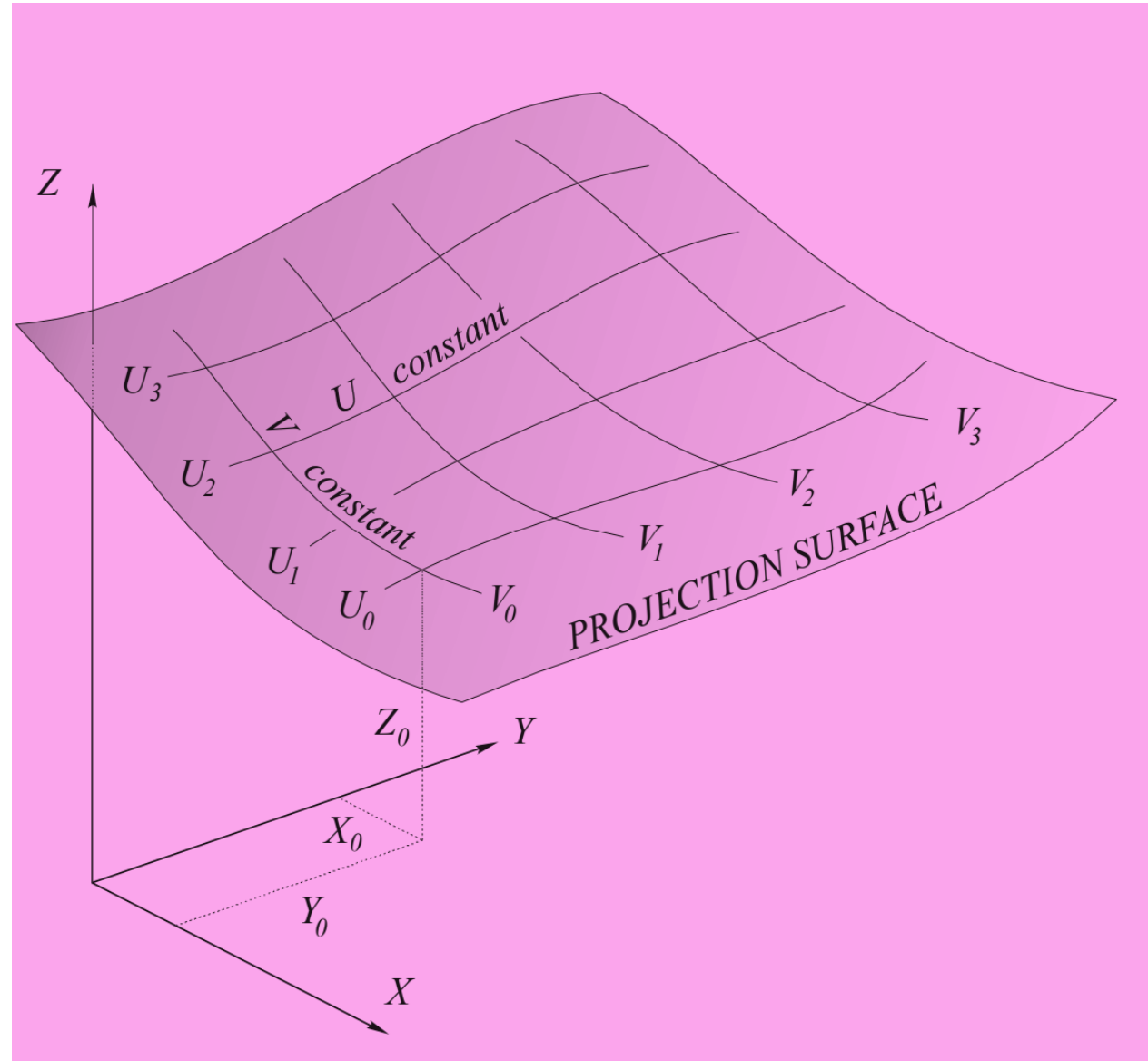


Projection Surface

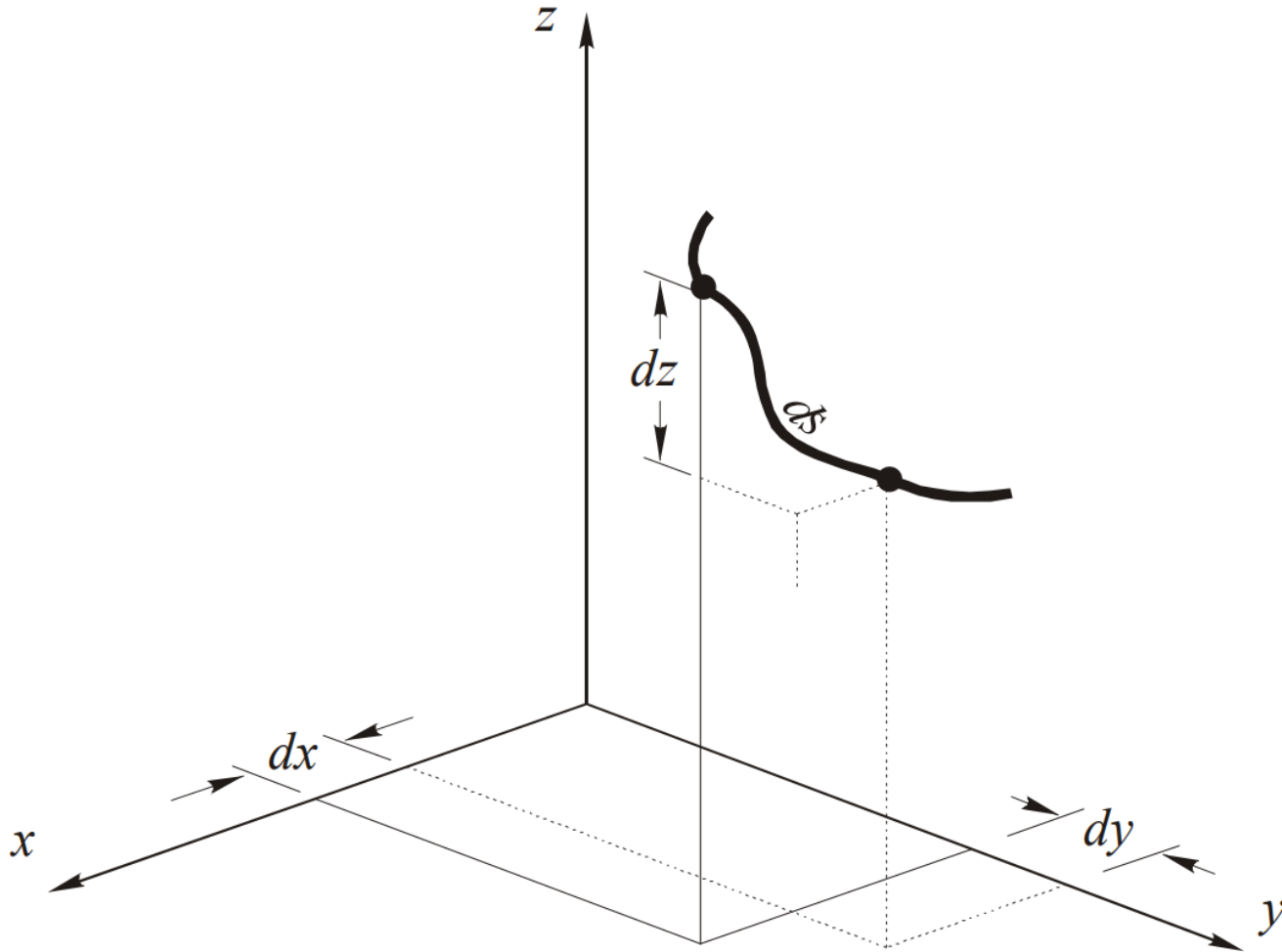
$$X = F_1(U, V)$$

$$Y = F_2(U, V)$$

$$Z = F_3(U, V)$$



THE GAUSSIAN FUNDAMENTAL QUANTITIES ON THE DATUM SURFACE



From differential geometry, the square of the length of a differentially small part of a curve on the datum surface is

$$ds^2 = dx^2 + dy^2 + dz^2 \quad (3.1)$$

$$dx = \frac{\partial x}{\partial u} du + \frac{\partial x}{\partial v} dv$$

$$dy = \frac{\partial y}{\partial u} du + \frac{\partial y}{\partial v} dv$$

$$dz = \frac{\partial z}{\partial u} du + \frac{\partial z}{\partial v} dv$$

$$ds^2 = \left(\frac{\partial x}{\partial u} \right)^2 du^2 + \left(\frac{\partial x}{\partial v} \right)^2 dv^2 + 2 \frac{\partial x}{\partial u} \frac{\partial x}{\partial v} du dv$$

$$+ \left(\frac{\partial y}{\partial u} \right)^2 du^2 + \left(\frac{\partial y}{\partial v} \right)^2 dv^2 + 2 \frac{\partial y}{\partial u} \frac{\partial y}{\partial v} du dv$$

$$+ \left(\frac{\partial z}{\partial u} \right)^2 du^2 + \left(\frac{\partial z}{\partial v} \right)^2 dv^2 + 2 \frac{\partial z}{\partial u} \frac{\partial z}{\partial v} du dv$$

$$\begin{aligned} ds^2 = & \left\{ \left(\frac{\partial x}{\partial u} \right)^2 + \left(\frac{\partial y}{\partial u} \right)^2 + \left(\frac{\partial z}{\partial u} \right)^2 \right\} du^2 \\ & + \left\{ \frac{\partial x}{\partial u} \frac{\partial x}{\partial v} + \frac{\partial y}{\partial u} \frac{\partial y}{\partial v} + \frac{\partial z}{\partial u} \frac{\partial z}{\partial v} \right\} 2 du dv \\ & + \left\{ \left(\frac{\partial x}{\partial v} \right)^2 + \left(\frac{\partial y}{\partial v} \right)^2 + \left(\frac{\partial z}{\partial v} \right)^2 \right\} dv^2 \end{aligned}$$

Gaussian Fundamental Quantities

$$ds^2 = e du^2 + 2f du dv + g dv^2$$

$$e = \left(\frac{\partial x}{\partial u} \right)^2 + \left(\frac{\partial y}{\partial u} \right)^2 + \left(\frac{\partial z}{\partial u} \right)^2$$

$$f = \frac{\partial x}{\partial u} \frac{\partial x}{\partial v} + \frac{\partial y}{\partial u} \frac{\partial y}{\partial v} + \frac{\partial z}{\partial u} \frac{\partial z}{\partial v}$$

$$g = \left(\frac{\partial x}{\partial v} \right)^2 + \left(\frac{\partial y}{\partial v} \right)^2 + \left(\frac{\partial z}{\partial v} \right)^2$$

Elemental distance on the surface of the spherical Earth

$$x = f_1(\phi, \lambda) = R \cos \phi \cos \lambda$$

$$y = f_2(\phi, \lambda) = R \cos \phi \sin \lambda$$

$$z = f_3(\phi, \lambda) = R \sin \phi$$

$$e = \left(\frac{\partial x}{\partial \phi} \right)^2 + \left(\frac{\partial y}{\partial \phi} \right)^2 + \left(\frac{\partial z}{\partial \phi} \right)^2$$

$$f = \frac{\partial x}{\partial \phi} \frac{\partial x}{\partial \lambda} + \frac{\partial y}{\partial \phi} \frac{\partial y}{\partial \lambda} + \frac{\partial z}{\partial \phi} \frac{\partial z}{\partial \lambda}$$

$$g = \left(\frac{\partial x}{\partial \lambda} \right)^2 + \left(\frac{\partial y}{\partial \lambda} \right)^2 + \left(\frac{\partial z}{\partial \lambda} \right)^2$$

Gaussian Fundamental Quantities for Sphere

$$\frac{\partial x}{\partial \phi} = -R \sin \phi \cos \lambda$$

$$\frac{\partial y}{\partial \phi} = -R \sin \phi \sin \lambda$$

$$\frac{\partial x}{\partial \phi} = R \cos \phi$$

$$\frac{\partial x}{\partial \lambda} = -R \cos \phi \sin \lambda$$

$$\frac{\partial x}{\partial \lambda} = R \cos \phi \cos \lambda$$

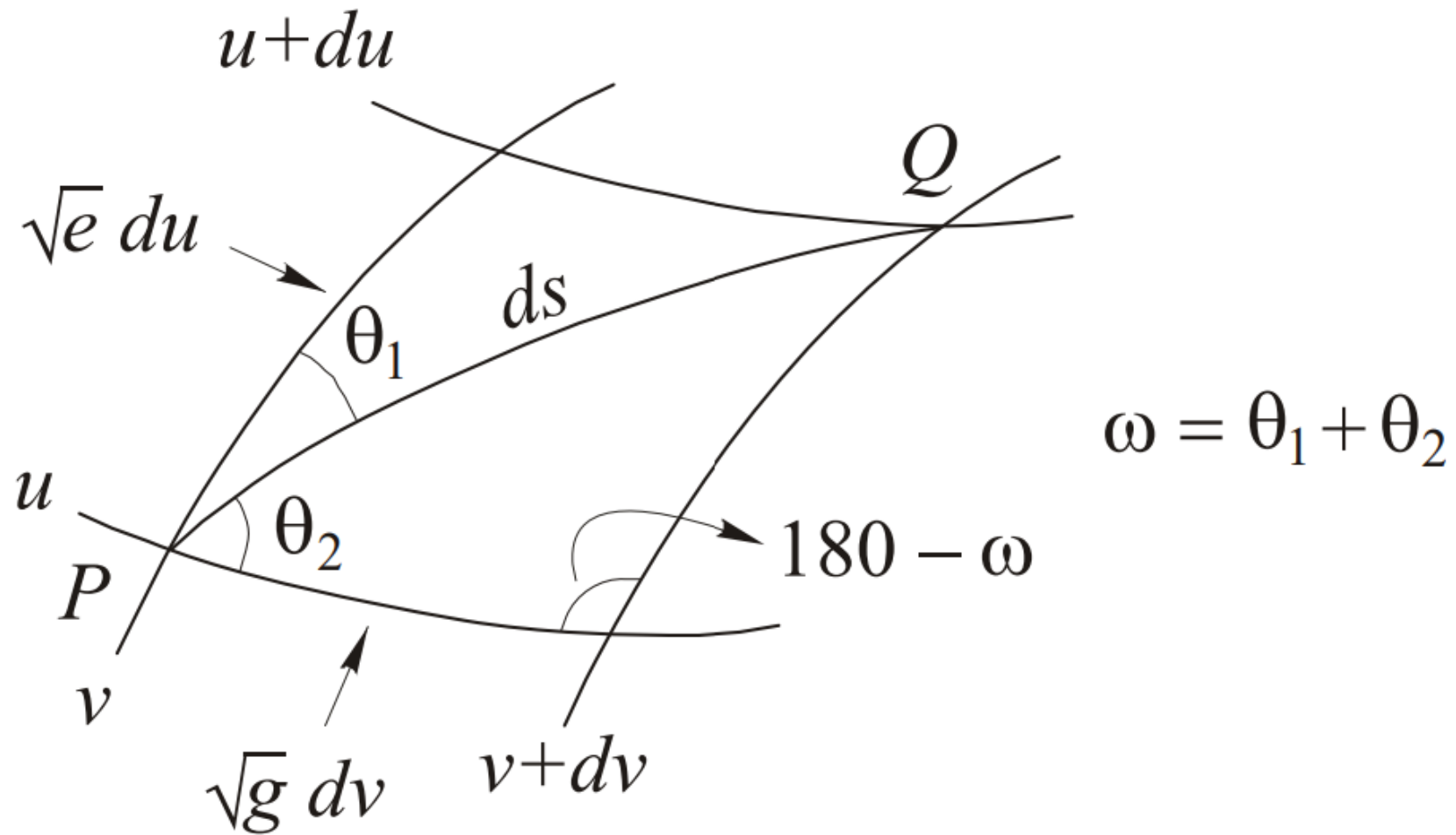
$$\frac{\partial x}{\partial \lambda} = 0$$

$$e = R^2$$

$$f = 0$$

$$g = R^2 \cos^2 \phi$$

PARALLELOGRAM ON THE DATUM SURFACE



distance along parametric curves

$$\begin{aligned} ds_v^2 &= e du^2 + 2f du dv + g dv^2 \\ &= e du^2 + 2f du(0) + g(0)^2 \\ &= e du^2 \end{aligned}$$

$$ds_v = \sqrt{e} du$$

$$\begin{aligned} ds_u^2 &= e du^2 + 2f du dv + g dv^2 \\ &= e(0)^2 + 2f(0)dv + g dv^2 \\ &= g dv^2 \end{aligned}$$

$$ds_u = \sqrt{g} dv$$

angle between parametric curves on the datum surface

$$\begin{aligned} ds^2 &= e du^2 + g dv^2 - 2(\sqrt{e} du)(\sqrt{g} dv)\cos(180 - \omega) \\ &= e du^2 + g dv^2 + 2\sqrt{eg} du dv \cos \omega \end{aligned}$$

$$\cos \omega = \frac{f}{\sqrt{eg}}$$

$$\sin \omega = \frac{j}{\sqrt{eg}}$$

area on the datum surface

$$\begin{aligned} da &= \left(\sqrt{e} du \right) \left(\sqrt{g} dv \right) \sin \omega \\ &= \sqrt{eg} \sin \omega du dv \end{aligned}$$

$$da = j du dv$$

- Note:
- (i) The Gaussian Fundamental Quantities pertaining to the datum surface (x, y, z as functions of u and v) are denoted by e, f and g .
 - (ii) The Gaussian Fundamental Quantities pertaining to the projection surface (X, Y, Z as functions of U and V) are denoted by $\bar{E}, \bar{F}$ and $\bar{G}$.
 - (iii) The Gaussian Fundamental Quantities pertaining to the projection surface (X, Y, Z as functions of u and v) are denoted by E, F and G .

